End-to-End McKinsey Enterprise Value Creation, Capital Allocation & AI CFO Intelligence Platform

Finance Analytics Report

MS SQL Server | Python | Power BI | Corporate Finance

Project objective
Build an explainable CFO decision-support platform that identifies where the enterprise creates or destroys value, forecasts performance, allocates scarce capital, and prioritizes executive actions.

Portfolio disclaimer
Independent synthetic portfolio project inspired by publicly available corporate-finance and value-creation practices. It is not affiliated with, endorsed by, commissioned by, or representative of confidential work performed by McKinsey & Company. All company and financial data are synthetic.

Prepared for portfolio / GitHub / resume use

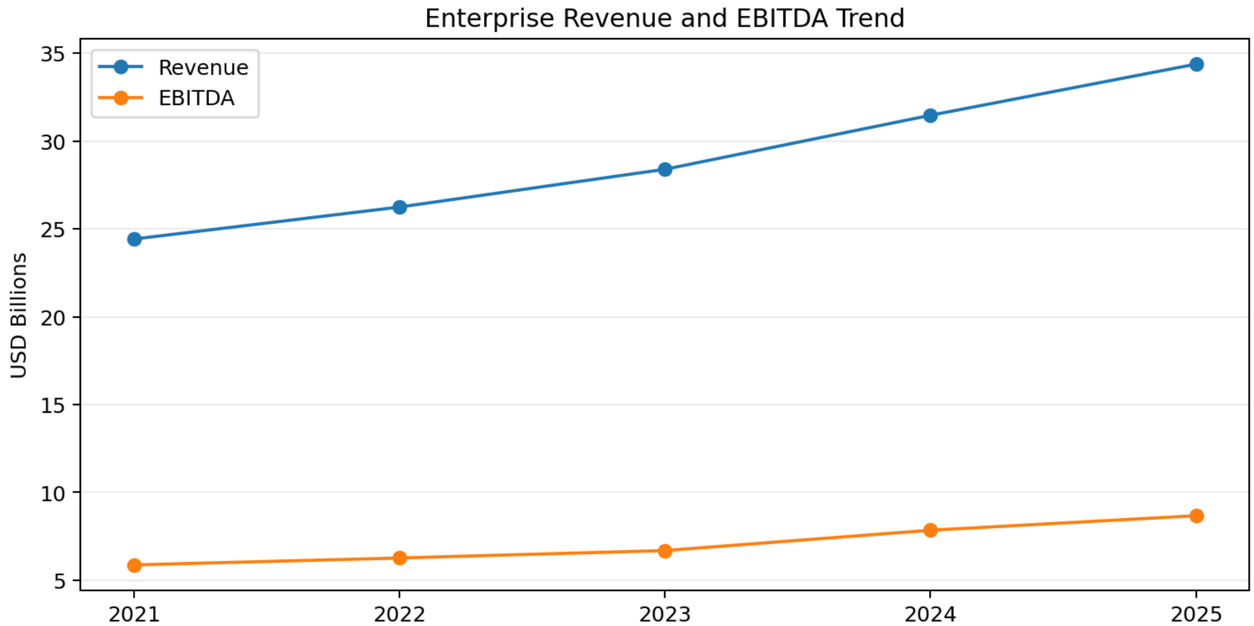
1. Executive Summary

Apex Global Holdings is a fictional diversified enterprise analyzed through an end-to-end finance analytics platform. The solution integrates operating performance, value creation, planning, forecasting, capital allocation, and explainable CFO intelligence in one reproducible workflow.

KPI	Validated Result
2025 Revenue	$34.38B
2025 EBITDA	$8.67B
2025 EBITDA Margin	25.2%
2025 Free Cash Flow	$4.19B
Enterprise ROIC	19.2%
WACC	9.9%
ROIC - WACC Spread	+9.3 pp
Economic Profit	~$2.50B
Net Debt / EBITDA	3.07x
2026 Capital Requested	$2.676B
2026 Base Capital Budget	$1.650B
Optimized Portfolio NPV	~$494.84M

Most important finding

Industrial Systems is the clearest turnaround case: approximately 4.7% ROIC versus 9.9% WACC, a negative spread of about 5.2 percentage points, and approximately -$557M economic profit at the latest historical period.


Enterprise Revenue and EBITDA Trend
Revenue
EBITDA
USD Billions
35
30
25
20
15
10
5
2021
2022
2023
2024
2025

2. Business Problem & Solution Architecture

The platform answers four linked CFO questions:

- Where is financial performance improving or deteriorating?
- Which businesses create value above the cost of capital, and which destroy value?
- What is likely to happen next versus budget?
- Which investments should receive scarce capital, and what should the CFO act on first?

End-to-end workflow

Stage	Output
Raw data	6 synthetic CSV datasets with controlled data-quality issues
MS SQL Server	Raw relational source tables
Python validation & cleaning	Clean analytical layer with preserved source history
Feature engineering	EBITDA, NOPAT, FCF, ROIC, WACC spread, Economic Profit, leverage
SQL analytics	Financial, value-creation, budget, capital and CFO views
Forecasting	2026 Revenue, EBITDA and FCF forecasts with MAPE back-testing
Capital allocation	NPV/IRR/ROI/payback scoring and constrained portfolio optimization
AI CFO intelligence	Prioritized, explainable risks, opportunities and actions
Power BI	5-page executive CFO dashboard

Design principle

The LLM is not the source of truth for finance. Financial calculations are deterministic and auditable; the intelligence layer only prioritizes verified metrics and converts them into executive actions.

3. Data Design, Validation & Cleaning

The synthetic dataset covers January 2021 through December 2025 across five business units and four regions, with budget data extending through 2026 and scenario assumptions through 2030.

Dataset	Clean rows	Primary use
business_units	5	Business-unit dimension
financial_performance	1,200	Monthly P&L and reinvestment drivers
balance_sheet	1,200	Working capital, fixed assets, cash and debt
budget_forecast	720	Budget variance and forward plan
capital_projects	40	Investment appraisal and optimization
market_assumptions	20	WACC, macro and Bear/Base/Bull scenarios

Controlled raw-data imperfections

- 6 financial-performance duplicate rows and 10 missing numeric cells.
- 4 balance-sheet duplicate rows and 8 missing numeric cells.
- 3 budget duplicate rows and 6 missing numeric cells.
- No corrupted business-unit keys, invalid dates, broken region relationships, or invalid WACC/terminal-growth relationships.

Cleaning strategy

- Remove exact duplicates while leaving the raw SQL source unchanged.
- Fill missing numeric values using same-BU / same-region time-series interpolation, then BU median fallback.
- Validate dates, data types, rate ranges and non-negative operating values.
- Flag unusual one-off observations instead of blindly winsorizing meaningful business events.

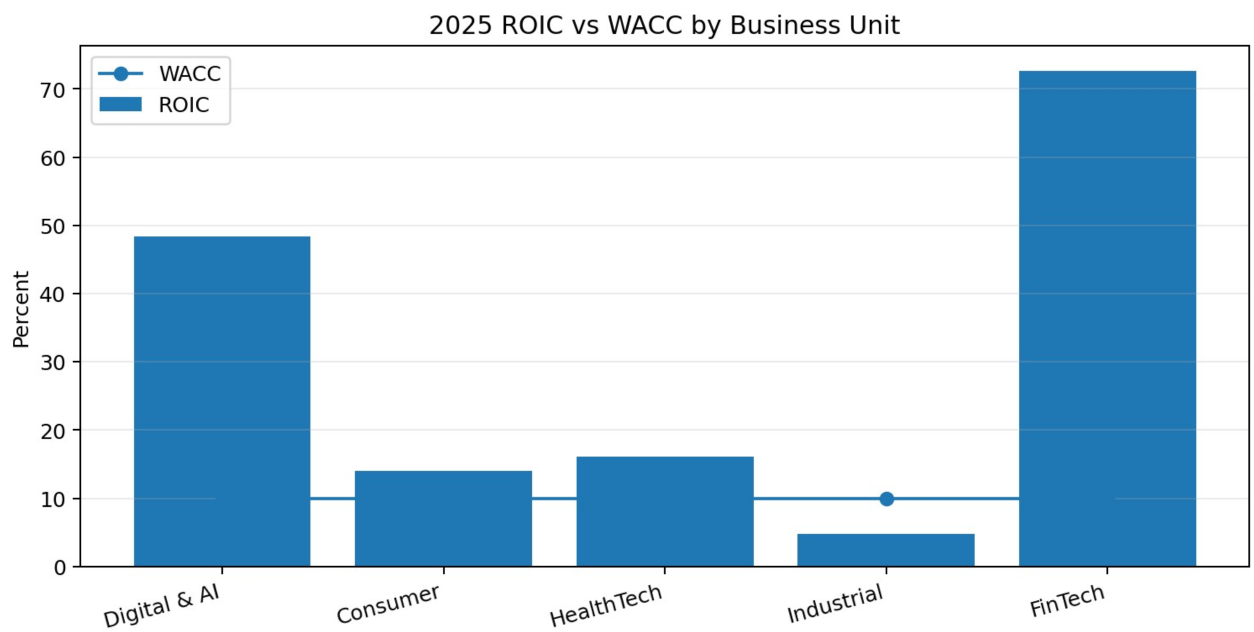
Why this matters

The pipeline deliberately preserves genuine signals such as Industrial Systems capital intensity and FinTech investment volatility. Cleaning improves data quality without erasing the business story.

4. Financial Performance & Enterprise Value Creation

The analytical layer converts raw operating and balance-sheet drivers into CFO metrics including EBITDA, EBIT, NOPAT, Free Cash Flow, Invested Capital, Net Debt, ROIC, ROIC-WACC spread, Economic Profit and leverage.

Formula	Definition
EBITDA	Revenue - COGS - Operating Expenses
NOPAT	EBIT x (1 - Effective Tax Rate)
Free Cash Flow	NOPAT + D&A - CAPEX - Change in NWC
Net Working Capital	Receivables + Inventory - Payables
Invested Capital	Net Working Capital + Net Fixed Assets
ROIC	TTM NOPAT / Average Invested Capital
Economic Profit	TTM NOPAT - (Average Invested Capital x WACC)



Business Unit	ROIC	WACC	Spread	Economic Profit
Digital & AI Services	48.3%	9.9%	+38.4 pp	$1,429M
Consumer Brands	14.0%	9.9%	+4.1 pp	$223M
HealthTech Solutions	16.1%	9.9%	+6.2 pp	$331M
Industrial Systems	4.7%	9.9%	-5.2 pp	-$557M
FinTech Platforms	72.7%	9.9%	+62.8 pp	$1,071M

Interpretation note

Digital and FinTech show very high ROIC because the project uses a simplified operating invested-capital definition and both are relatively asset-light. The methodology is disclosed instead of silently capping the results.

5. Financial Forecasting

The forecasting layer predicts 2026 Revenue, EBITDA and Free Cash Flow at business-unit level. It is intentionally simple and explainable rather than using a model zoo.

Step	Design
Training period	January 2021 - December 2024
Back-test	January 2025 - December 2025
Models	12-month Moving Average vs Holt-Winters Exponential Smoothing
Selection metric	MAPE
Final forecast horizon	January 2026 - December 2026
Forecast grain	Month x Business Unit

Metric	Average Back-test Accuracy	Interpretation
Revenue	97.1%	Strong and stable trend/seasonality capture
EBITDA	93.8%	Strong operating-profit forecastability
Free Cash Flow	48.3%	Material volatility from CAPEX and working capital

CFO insight

Weak FCF forecast accuracy is not hidden. It becomes an explicit planning-risk signal, highlighting where cash forecasting requires more active monitoring and scenario management.

Example 2026 forecast-vs-budget signals

Business Unit	Revenue vs Budget	EBITDA vs Budget
Digital & AI Services	+2.8%	-1.0%
Consumer Brands	+1.6%	-6.5%
HealthTech Solutions	-1.4%	-1.8%
Industrial Systems	-3.7%	-8.7%
FinTech Platforms	+3.4%	+0.3%

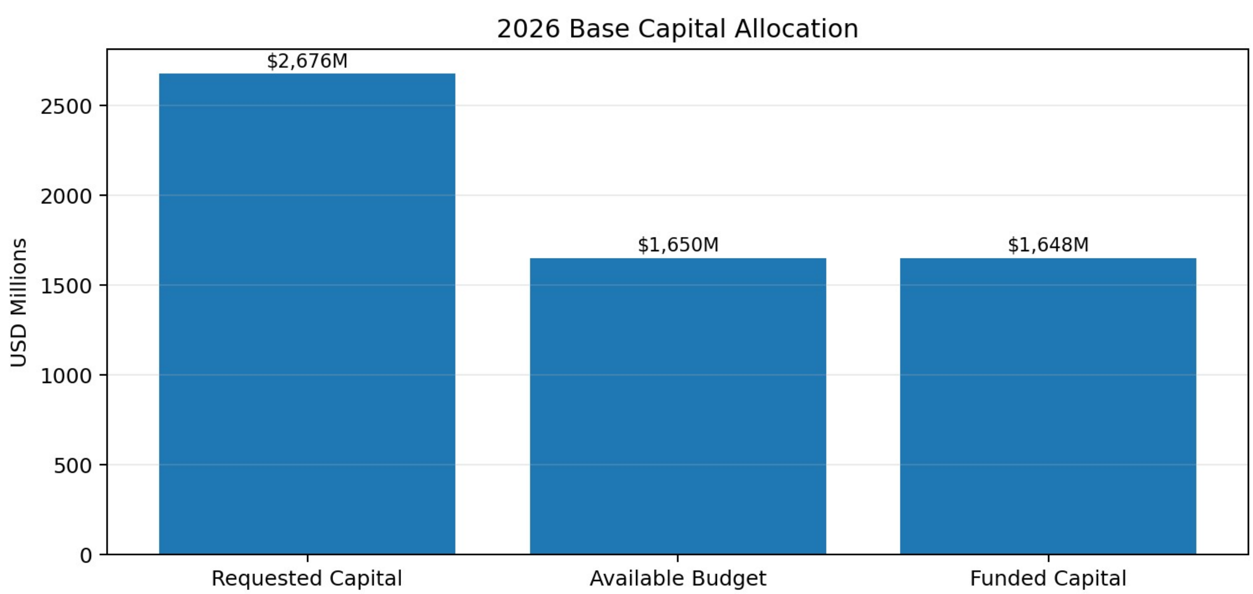
6. Capital Allocation Engine

The project evaluates 40 proposed investments using project economics, strategic priority and risk, then solves the portfolio-level capital constraint.

Metric	Weight
Normalized NPV	35%
Normalized IRR	20%
Normalized ROI	10%
Normalized Payback	10%
Strategic Priority	15%
Inverse Risk	10%

Finance rule

NPV remains primary. The 0-100 Capital Allocation Score is a ranking aid for management interpretation; it does not override the requirement to create economic value and clear the WACC hurdle.



Portfolio Result	Value
Projects evaluated	40

Positive-NPV / hurdle-clearing projects	27
Requested capital	$2.676B
2026 Base capital budget	$1.650B
Funding gap	$1.026B
Projects selected	22
Capital deployed	$1.648B
Unused capital	$2M
Optimized portfolio NPV	$494.84M

The optimization uses a transparent 0/1 dynamic-programming approach to maximize total positive NPV while staying within the available budget. The resulting portfolio uses virtually all available capital without funding projects that fail the economic hurdle.

7. AI CFO Intelligence Engine

The intelligence layer converts verified analytical outputs into a prioritized executive action queue. It is rule-based and fully auditable, with no mandatory paid LLM or external API.

CFO Priority component	Weight
Value-Destruction Severity	35%
EBITDA Budget-Miss Severity	25%
FCF Deterioration Severity	20%
Forecast Risk Severity	20%

Illustrative priority logic from the validated dataset

- Industrial Systems is the highest business-unit priority because ROIC is below WACC and EBITDA performance is materially below plan.
- Enterprise Net Debt / TTM EBITDA is approximately 3.07x, exceeding the project monitoring threshold of 3.0x.
- Capital demand exceeds the Base budget by $1.026B, so lower-value projects must be deferred.
- Top funded investments are surfaced as opportunities, not only risks.

Explainability principle

Each action exposes the triggering finance metrics. The engine does not invent causes or recommendations that cannot be traced to the calculated data.

Example executive actions

Priority area	Recommended action
Industrial Systems	Improve margins and capital efficiency before major expansion; review cost drivers and discretionary CAPEX.
Leverage	Protect cash generation and limit non-priority debt-funded investment.
Capital allocation	Fund the optimized positive-NPV portfolio and defer lower-value projects.
Forecast risk	Refresh plan assumptions where forecast accuracy or budget risk is weak.

8. Power BI Executive Dashboard

The final report deliberately uses five pages to balance simplicity, quality and executive storytelling. Each page answers one distinct CFO question.

Page	Purpose	Key visuals
1. Executive CFO Overview	How is the enterprise performing?	KPI strip, Revenue/EBITDA trend, BU revenue, top CFO priorities
2. Enterprise Value Creation	Where is value created or destroyed?	ROIC vs WACC, Economic Profit, Growth vs Return, scorecard
3. Budget, Forecast & Scenario	Where are plan risks and what happens next?	Budget variance, 2026 forecast, accuracy, Bear/Base/Bull assumptions
4. Capital Allocation Optimizer	Where should scarce capital go?	Capital KPIs, NPV scatter, funded projects, decision table
5. AI CFO Intelligence	What should the CFO act on first?	Priority scores, action queue, selected decision detail

Screenshot naming standard

01_executive_cfo_overview.png | 02_enterprise_value_creation.png | 03_budget_forecast_scenario_intelligence.png | 04_capital_allocation_optimizer.png | 05_ai_cfo_intelligence.png

The Power BI model uses Import mode, a dedicated Date table, a Business Unit dimension, single-direction relationships, and a dedicated measures table. This keeps the report fast, portable and understandable for portfolio reviewers.

9. Technical Implementation & Portfolio Value

Layer	Technology
Database	Microsoft SQL Server
Data processing	Python - pandas / NumPy
SQL connectivity	SQLAlchemy / pyodbc
Forecasting	statsmodels
Investment analytics	numpy-financial
Optimization	Python dynamic programming
Dashboard	Microsoft Power BI / DAX

What the project demonstrates

- End-to-end finance analytics architecture from raw data to executive dashboard.
- Corporate-finance fluency beyond standard revenue/profit reporting.
- Transparent data-quality, forecasting and model-evaluation logic.
- Real capital-allocation trade-offs under a binding budget constraint.
- Explainable decision intelligence rather than opaque black-box recommendations.
- A recruiter-friendly Power BI narrative with only five focused pages.

Conclusion

The platform connects operating performance, economic value creation, forecasting, investment decisions and executive prioritization into one coherent CFO workflow. Its strongest feature is not any single algorithm; it is the traceable link from raw financial drivers to a management decision.

Final portfolio message

Where is value being created, where is it being destroyed, what happens next, where should the next dollar of capital go, and what should the CFO do first? This project is designed to answer all five questions in one system.

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